When Money Aggregates Failed Why the Money “Abandoned” Us

Susanna Grigoryan

Recent Staff Member, Monetary Policy Department, Central Bank of Armenia

Student, Global Forecasting School

November 2025

Note

Monetary targeting emerged as a leading policy framework in the late 1970s and early 1980s as central banks searched for a clearer and more disciplined nominal anchor following the Great Inflation. The foundational logic came from the monetarist proposition that inflation is fundamentally the result of excessive money growth. If the central bank could restrain the growth rate of monetary aggregates to a predictable path, inflation would, over time, converge to a stable rate. This note explains the appeal of monetary targeting, the operational challenges that quickly emerged, and the lessons for good economists.

1. Logic and Appeal of Monetary Targets

The basic framework rested on three assumptions:

1. money demand was stable
2. velocity changed only gradually, and
3. output followed a predictable trend.

Under these conditions, targeting the growth rate of aggregates such as M1 or M2 seemed like a reliable method for anchoring inflation. Central banks in advanced economies, including the United States, Canada, the United Kingdom, and Germany, adopted explicit targets for money growth. Armenia, like many other countries transitioning toward modern frameworks, later experimented with similar approaches.

2. The CBA’s experience with Money Aggregates

CENTRAL BANK
OF ARMENIA

THE BETTER POLICY PROJECT
GREAT THINGS ARE COMING

From 1994 to 2005, the Central Bank of Armenia operated under a **monetary aggregate targeting** framework. This approach was formalized with the 1996 *Law on the Central Bank of Armenia*, which established price stability as the primary policy objective, restricted direct lending to the government, mandated the use of indirect instruments, and strengthened central bank independence.

Broad money served as the **intermediate target**, while the **monetary base** operated as the **operational target**. The CBA influenced banking system liquidity and banks' investment decisions through a mix of liquidity-providing (Lombard loans, repos, open-market and FX purchases) and liquidity-absorbing (deposit auctions, repos, open-market and FX sales) instruments. The goal was to steer monetary aggregates toward planned trajectories consistent with achieving low inflation.

The regime was initially successful. Inflation, though extremely high in 1994 (1,761% year-on-year), became manageable in the second half of that year. By 1995, inflation had fallen to 32.2%, and real GDP returned to positive growth. From 1995 to 1998, inflation stabilized, and Armenia entered a low-inflation environment accompanied by rapid economic growth, including consecutive double-digit expansions from 2003 to 2007.

However, the framework soon faced structural difficulties. Until 1998, broad money (M2X)—including foreign-currency deposits—served as the nominal anchor. As dollarization increased, targeting broad money constrained growth in dram-denominated liquidity, risking a contraction in aggregate demand. To address this, the CBA began targeting dram-based aggregates in 1998, though this did not resolve inconsistencies between the operational target and the nominal anchor.

Growing volatility in money demand, the shadow circulation of U.S. dollars, rising external inflows after 2000, and a weak transmission mechanism undermined the stability of both the monetary base–broad money relationship and the broad money–inflation link. Broad money persistently deviated from targets, forcing frequent revisions and eroding the credibility of the nominal anchor. With broad money no longer a reliable anchor, the CBA faced a strategic choice between adopting an exchange-rate anchor or transitioning to inflation targeting.

CENTRAL BANK
OF ARMENIA

THE BETTER POLICY PROJECT
GREAT THINGS ARE COMING

3. Early Operational Difficulties

Almost immediately, policymakers encountered large and persistent deviations of monetary aggregates from their intended targets. These deviations often carried ambiguous implications for inflation. In some episodes, rapid money growth did not translate into higher inflation; in others, inflation rose even when monetary aggregates appeared to be under control. The once-stable empirical relationship between money and nominal income began to deteriorate.

4. Structural Breaks in Money Demand

Financial innovation played a central role in destabilizing previously stable money-demand functions. Deregulation in the banking sector, the introduction of interest-bearing deposits, the rise of money market mutual funds, and technological changes in payments all altered how households and firms managed liquidity. As a result, traditional aggregates like M1 and M2 ceased to track the underlying liquidity positions of the private sector. Velocity became unpredictable, and former models no longer provided reliable guidance.

A good economist recognizes these structural breaks as soon as they emerge. When a key empirical relationship becomes unstable, it cannot serve as a reliable operational foundation for policy.

5. Endogenous Money and Banking Behaviour

A deeper conceptual issue was that monetary targeting did not reflect the endogenous nature of modern money creation. When banks expand credit, deposits rise; when banks contract lending, deposits fall. These dynamics depend more on economic conditions and risk appetite than on central bank actions. Monetary aggregates therefore reflected the behaviour of private financial institutions rather than the intended stance of monetary policy. Attempts to control the quantity of money led to volatility in short-term interest rates and credit conditions, raising concerns about financial stability. Central banks were forced to choose between stabilizing interest rates or stabilizing money growth—most chose the former.

Recent research has begun filling a gap in how economists model these mechanisms. For example, DSGE models with explicit banking, balance sheet interactions, and liquidity and maturity transformations such as Benes, Kumhof, and Laxton (2014) and Mkhatrishvili (2025)—provide modern treatments of endogenous money creation. These papers are not about monetary targeting itself; rather, they offer improved analytical foundations for thinking about money, credit, and liquidity inside structural policy models.

6. Transition to Interest-Rate Instruments

As practical challenges mounted, central banks shifted from quantity-based to price-based operational frameworks. Instead of attempting to control monetary aggregates directly, policymakers began to influence economic activity and inflation by setting a Short-term policy interest rate and shaping expectations about its future path. This approach proved more effective, more flexible, and more consistent with the functioning of modern financial systems.

For Armenia, as in many other countries, experiences with monetary targeting provided valuable lessons that ultimately led to the adoption of inflation targeting frameworks supported by forecasting and policy analysis systems.

7. Lessons for Good Policy Analysis

The monetary targeting era taught economists several important lessons:

- Empirical relationships must be continuously evaluated for stability.
- Structural change can disrupt long-standing regularities.
- Monetary and credit aggregates are useful indicators but cannot serve as the sole anchor of policy.
- Good economists adapt frameworks when evidence shows that earlier assumptions no longer hold.

These insights paved the way for modern inflation-targeting regimes and the development of FPAS-type frameworks, while recent work on endogenous money creation helps rebuild the analytical foundations for thinking about money, credit, and liquidity inside those frameworks.

CENTRAL BANK OF ARMENIA

THE BETTER POLICY PROJECT
GREAT THINGS ARE COMING

Literature & Further Reading

- Benes, J., M. Kumhof, and D. Laxton. 2014. Financial Crises in DSGE Models: Selected Applications of MAPMOD. IMF Working Paper WP/14/56. International Monetary Fund.
- Brunner, K., and A. Meltzer. 1993. Money and the Economy. Cambridge University Press.
- Friedman, M. 1968. "The Role of Monetary Policy." *American Economic Review* 58 (1).
- Goodhart, C. 1984. Monetary Theory and Practice: The UK Experience. Macmillan.
- Laidler, D. 1999. The Quantity of Money and Monetary Policy. Bank of Canada Review.
- Laxton, D., M. Galstyan, and V. Avagyan (eds.). 2023. History of Monetary Policy in Armenia. The Better Policy Project. Unpublished mimeograph.
- Mkhatrishvili, S. 2025. Inside (the) Money Machine: Modeling Liquidity, Maturity and Credit Transformations. IMF Working Paper WP/25/166. International Monetary Fund.
- Laxton, D., Avetisyan, H. and Avagyan, V.. December 2025. Prudent Risk Management Approach to Price Stability: Theory and Practice of FPAS Mark II at the Central Bank of Armenia. Unpublished Working Draft.

Script

Hello, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level 1 student at the Global Forecasting School**. In this video, I will explain why many central banks in the 1970s and early 1980s attempted to control inflation by targeting the growth rate of money.

After the Great Inflation, policymakers wanted a clear and rule-based way to bring inflation down. Milton Friedman's idea that inflation is always and everywhere a monetary phenomenon became extremely influential. If inflation results from excessive money growth, then controlling the growth of monetary aggregates like M1 or M2 should allow the central bank to control inflation.

Central banks across the world adopted monetary targets. They announced desired paths for money growth and adjusted their policy instruments to try to keep actual money growth in line with those targets. In the early stages, this approach appeared promising. But problems quickly emerged.

Monetary aggregates often behaved unpredictably. Sometimes money growth rose sharply without producing much inflation; in other periods, inflation increased even though money growth seemed under control. These inconsistencies reflected deeper structural changes happening in the financial system.

Financial innovation — including deregulation, new types of deposits, and improved payments technology — altered how households and firms held liquid assets. Traditional measures of money no longer captured the concept reliably. At the same time, central banks discovered that they did not directly control the money supply: in modern economies, most money is created by banks through lending. When credit expanded, money grew; when credit contracted, money slowed — regardless of the central bank's intentions.

A good economist learns from these developments. When an indicator stops providing reliable information, it cannot remain the centre of a policy framework. These experiences led central banks to shift from controlling the quantity of money to influencing the price of money — the Short-term policy interest rate. This approach proved far more effective and flexible. For Armenia, as for many other countries, monetary targeting served as a transitional phase on the path toward inflation targeting supported by forecasting and policy analysis systems.

I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance and support in **preparing this material.**